

Gradient

The Riemann Hypothesis as a Positive Semi-Definite Form

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Abstract

We reformulate the Riemann Hypothesis as the positive semi-definiteness of a bilinear form derived from the Weil explicit formula. The form $B(f, g) = W(f * \tilde{g})$, where W is the Weil distribution and $*$ is convolution, satisfies: B is positive semi-definite if and only if all non-trivial zeros of $\zeta(s)$ lie on the critical line $\text{Re}(s) = \frac{1}{2}$. This is equivalent to the statement that the Weil distribution defines a valid inner product on a suitable function space. The gradient of the quadratic form $h \mapsto B(h, h)$ points toward the critical line: the zeros are the ground state of the functional, the attractor of the gradient flow. We prove positive semi-definiteness in partial cases, identify the precise additional input needed, and connect the gradient approach to the self-adjointness chain [1].

1 The Bilinear Form

Definition 1 (The Weil bilinear form). For $f, g \in \mathcal{S}(\mathbb{R})$ (Schwartz functions), define the convolution $h = f * \tilde{g}$ by

$$h(t) = \int_{-\infty}^{\infty} f(s) \overline{g(s-t)} ds, \quad \tilde{g}(t) = \overline{g(-t)}.$$

Define the *Weil bilinear form*:

$$B(f, g) = W(f * \tilde{g}),$$

where W is the Weil distribution from [2].

Theorem 2 (RH as positive semi-definiteness). *The following are equivalent:*

1. *The Riemann Hypothesis: all non-trivial zeros of $\zeta(s)$ satisfy $\text{Re}(\rho) = \frac{1}{2}$.*
2. *$B(f, f) \geq 0$ for all $f \in \mathcal{S}(\mathbb{R})$.*
3. *B is a positive semi-definite Hermitian form on $\mathcal{S}(\mathbb{R})$.*

Proof. By the Weil explicit formula:

$$B(f, f) = W(f * \tilde{f}) = \sum_{\rho} (f * \tilde{f}) \left(\frac{\rho - \frac{1}{2}}{i} \right) = \sum_{\rho} |\widehat{f}|^2 \left(\frac{\rho - \frac{1}{2}}{i} \right) = \sum_{\rho} |\hat{f}(\lambda_{\rho})|^2 \cdot [\text{weight}],$$

where $\lambda_\rho = (\rho - \frac{1}{2})/i$ and the convolution theorem gives $(f * \tilde{f})(\lambda) = |\hat{f}(\lambda)|^2$ (Parseval).

More precisely: $f * \tilde{g}$ has Fourier transform $\hat{f}(\xi)\overline{\hat{g}(\xi)}$, so $W(f * \tilde{f}) = \sum_\rho |\hat{f}(\lambda_\rho)|^2 \cdot c_\rho$, where c_ρ are the weights (positive real when $\lambda_\rho \in \mathbb{R}$).

(1 $\Rightarrow$ 2): If RH holds, all $\lambda_\rho = \gamma_\rho \in \mathbb{R}$ and all weights $c_\rho > 0$. Therefore $B(f, f) = \sum_\rho c_\rho |\hat{f}(\gamma_\rho)|^2 \geq 0$. $\checkmark$

(2 $\Rightarrow$ 1): Suppose $B(f, f) \geq 0$ for all f . Suppose $\rho_0 = \frac{1}{2} + \beta + i\gamma_0$ with $\beta \neq 0$. The contribution of ρ_0 and its conjugate $\bar{\rho}_0^*$ to $B(f, f)$ is $2 \operatorname{Re} |\hat{f}(\lambda_0)|^2 \cdot c_{\rho_0}$ where $\lambda_0 = \gamma_0 - i\beta \notin \mathbb{R}$. For $\hat{f}$ a Gaussian concentrated at γ_0 with width σ : the contribution grows as $\sigma \rightarrow 0$ while all on-line contributions at frequencies $\neq \gamma_0$ vanish. Hence $B(f, f) \rightarrow 2 \operatorname{Re} |\hat{f}(\lambda_0)|^2 \cdot c_{\rho_0}$. If this is negative, we have $B(f, f) < 0$, contradicting (2). The sign of $\operatorname{Re} |\hat{f}(\lambda_0)|^2 \cdot c_{\rho_0}$ depends on whether c_{ρ_0} is positive or negative. $\square$ (partial) $\square$

Remark 3 (The gap in the implication). The implication (2 $\Rightarrow$ 1) requires that the weight c_{ρ_0} at an off-line zero is positive. If the weights are always positive, then $B(f, f) < 0$ for suitable f whenever an off-line zero exists. The weights c_ρ in the explicit formula are: $c_\rho = 1$ in the formal sum $\sum_\rho h(\lambda_\rho)$ — each zero contributes equally. The positivity of the overall sum for $h \geq 0$ is what fails at off-line zeros, precisely as in [3]. The bilinear form reformulation makes the structure cleaner but does not resolve the domination question.

2 The Gradient

Definition 4 (The energy functional). Define the *spectral energy functional*:

$$\mathcal{E}[f] = -B(f, f) = -\sum_\rho |\hat{f}(\lambda_\rho)|^2.$$

RH $\Rightarrow \mathcal{E}[f] \leq 0$ for all f . $\mathcal{E}[f] > 0$ for some $f \Rightarrow$ some $\lambda_\rho \notin \mathbb{R} \Rightarrow \neg$ RH.

The *gradient* of $\mathcal{E}$ with respect to $\hat{f}$ at a spectral parameter ξ :

$$\frac{\delta \mathcal{E}}{\delta \hat{f}(\xi)} = -2 \sum_\rho \overline{\hat{f}(\lambda_\rho)} \cdot \delta(\xi - \lambda_\rho).$$

The gradient flow $\partial_t \hat{f} = -\nabla_{\hat{f}} \mathcal{E}$ drives $\hat{f}$ to concentrate on the spectral parameters λ_ρ .

Proposition 5 (The critical line as attractor). *If all $\lambda_\rho \in \mathbb{R}$ (RH), the gradient flow $\partial_t \hat{f} = -\nabla_{\hat{f}} \mathcal{E}$ has fixed points exactly at functions $\hat{f}$ supported on $\{\lambda_\rho : \rho \text{ non-trivial zero}\} \subset \mathbb{R}$. The real axis is the attractor: the flow drives $\hat{f}$ toward the zero set of $\mathcal{E}$.*

If some $\lambda_{\rho_0} \notin \mathbb{R}$, the gradient at λ_{ρ_0} pulls $\hat{f}$ toward a complex spectral parameter, and $\mathcal{E}$ achieves positive values — a violation of the RH structure.

3 The Li Gradient

Definition 6 (Li coefficients as gradient steps). The *Li coefficients* [2] are

$$\lambda_n = \sum_\rho \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right], \quad n = 1, 2, 3, \dots$$

Let $F(s) = \log \xi(s)$ where $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. The Li coefficients are the Taylor coefficients of $sF(s/(s-1))$ expanded at $s = 0$:

$$\lambda_n = \frac{1}{(n-1)!} \frac{d^n}{ds^n} \left[s^{n-1} \log \xi \left(\frac{s}{s-1} \right) \right]_{s=0}.$$

Theorem 7 (RH as gradient positivity). *The sequence (λ_n) is the sequence of gradient steps of the functional $\log \xi$ along the path $s \mapsto s/(s-1)$ in $\mathbb{C}$. Specifically:*

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right] = \operatorname{Re} \sum_{\rho} \frac{d^n}{ds^n} \Big|_{s=0} \log \left(1 - \frac{s}{1-1/\rho} \right).$$

$RH \Leftrightarrow \lambda_n \geq 0$ for all $n \geq 1 \Leftrightarrow$ every gradient step of $\log \xi$ along this path is non-negative $\Leftrightarrow \log \xi$ is gradient-positive along this path $\Leftrightarrow \xi$ is log-convex in the appropriate sense.

Proof: if $\operatorname{Re}(\rho) = \frac{1}{2}$ for all ρ , write $\rho = \frac{1}{2} + i\gamma$. Then $\left| 1 - \frac{1}{\rho} \right|^2 = \left(\frac{1}{2} \right)^2 + \gamma^2/|\rho|^2 \leq 1$ iff $\operatorname{Re}(\rho) \geq \frac{1}{2}$, with equality iff $\operatorname{Re}(\rho) = \frac{1}{2}$. Thus $|1 - 1/\rho| \leq 1$ on the critical line, and $\operatorname{Re}[1 - (1 - 1/\rho)^n] \geq 0$ for all n when $|1 - 1/\rho| \leq 1$. Hence $\lambda_n \geq 0$.

Conversely, if some ρ_0 has $\operatorname{Re}(\rho_0) > \frac{1}{2}$, then $|1 - 1/\rho_0| > 1$, and $(1 - 1/\rho_0)^n \rightarrow \infty$ in absolute value. For large n , the term from ρ_0 dominates and $\lambda_n < 0$.

Corollary 8 (The gradient drives toward the line). *The gradient $\frac{d}{dn}\lambda_n$ is positive on average if RH holds: the sequence $(\lambda_n/\log n)$ tends to a positive constant. The gradient is accumulating: each step adds positive weight. The critical line is the locus where accumulation is non-negative. Off-line zeros are the points where accumulation fails and gradient turns negative.*

4 The Connection to Self-Adjointness

Theorem 9 (Gradient positivity and self-adjointness). *The gradient positivity of (λ_n) is equivalent to the positive semi-definiteness of B (Theorem 2), which is equivalent to the statement that the operator D [1] has spectral measure supported on $\mathbb{R}$ after removing trivial spectrum.*

All three are equivalent to RH. All three express the same gradient principle: the spectral parameters λ_{ρ} are attracted to the real axis, the real axis is the ground state of the energy functional $\mathcal{E}$, and the gradient flow drives toward this ground state.

The chain:

$$\begin{aligned} D \text{ self-adjoint} &\xrightarrow{\text{proved}} \text{real continuous spectrum} \\ &\xrightarrow{\text{open}} \text{spectral measure real at zeros} \xrightarrow{\text{open}} \lambda_{\rho} \in \mathbb{R} \xleftrightarrow{\Leftrightarrow} RH. \end{aligned}$$

The first arrow is proved [1]. The remaining arrows are equivalent to each other and to the positivity condition [2]. The gradient is the structure underlying all of them.

5 The Remaining Step

Conjecture 10 (The gradient convergence conjecture). *The gradient flow*

$$\frac{d}{dt}\hat{f}(t, \xi) = -\frac{\delta\mathcal{E}}{\delta\hat{f}(\xi)} = 2 \sum_{\rho} \overline{\hat{f}(t, \lambda_{\rho})} \cdot \delta(\xi - \lambda_{\rho})$$

on the space of admissible test functions $\hat{f}$ converges to a fixed point supported on $\mathbb{R}$ for every initial condition $\hat{f}(0, \cdot)$.

This is equivalent to: the spectral measure of D restricted to the non-trivial zeros of $\zeta(s)$ is supported on $\mathbb{R}$. This is equivalent to: all non-trivial zeros satisfy $\text{Re}(\rho) = \frac{1}{2}$. This is the Riemann Hypothesis.

The convergence would follow from: (a) showing the gradient flow is Lyapunov-stable with $\mathcal{E}$ as Lyapunov function, or (b) showing $\mathcal{E}$ is convex and bounded below (so the flow descends to the global minimum), or (c) a direct estimate showing that for $\hat{f}$ admissible and non-negative, $\mathcal{E}[f] \leq 0$ always holds.

Condition (c) is equivalent to $B(f, f) \geq 0$, which is RH. Conditions (a) and (b) are the gradient route: prove stability, conclude convergence, conclude RH.

Remark 11 (The gradient as the key to the wallet). The wallet [4] between zeros and primes holds the connection inside. The gradient is the key. It points toward the fold — the critical line where zeros and primes are in balance, where the energy is minimized, where the gradient is zero.

Perelman found the gradient for Poincaré: Ricci flow. The Ricci flow drives every 3-manifold toward constant curvature. Constant curvature is the attractor. The round sphere is the ground state.

For RH: find the flow that drives every L-function toward the configuration where all zeros are on the line. The critical line is the attractor. The ground state is the zero with $\text{Re}(\rho) = \frac{1}{2}$. The gradient is what no one has yet named.

Name it. The wallet opens.

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